

8. Kidney transplants can save the lives of individuals with chronic kidney disease or kidney failure. In the UK in 2021, there were 2912 kidney transplants. Individuals who receive organ transplants must take drugs to suppress their immune response. However, it is estimated that 12.5% of kidneys transplanted are still rejected in the first year after a transplant.

- (a) Calculate the number of rejected kidney transplants in 2021. [2]
Space for working

number of rejected kidney transplants =

- (b) In 2019, it was reported that an individual who had received a kidney transplant had survived for 19 months without taking drugs to suppress their immune response. Before the transplant, this individual had a small number of stem cells taken from their bone marrow. The stem cells were grown and multiplied in the lab and then reinjected into the individual the day before the transplant. Some scientists suggested that the injection of stem cells from the bone marrow caused the body to accept the foreign antigens on the transplanted kidney.

Not all scientists were confident that the stem cells were responsible, because a second person given the same treatment was not able to survive without drugs to suppress their immune response.

- (i) State what is meant by the term **stem cell**. [2]

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- (ii) I. Name the type of cell division that occurs when stem cells are grown in the laboratory. [1]

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- II. Give **one** feature of this type of cell division which is important in the production of stem cells. [1]

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- (iii) State **one** benefit to the patient of using stem cells from their own bone marrow rather than stem cells from the bone marrow of another individual. [1]

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- (iv) Adult stem cells were used in this procedure. Name **one** other source of stem cells. [1]

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- (c) Suggest **one** way in which the scientific community could increase their confidence that the injection of stem cells was responsible for the result observed. [1]

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- (d) Dialysis is another treatment for kidney disease.
State **one** disadvantage of dialysis compared with kidney transplants. [1]

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